

T42 - Social Media Post Optimiser

AI-Powered Developer Tools Mini Project | Engagement Analysis and Posting Playbook

Dataset: 250 synthetic social media posts supplied in the assignment workbook. No external business data was used.

1. Executive Summary

Objective. The project analyses which post format, topic and posting time are associated with stronger engagement, uses an LLM to explain the patterns, verifies those explanations against spreadsheet calculations, generates 10 new posts, and converts the findings into a posting-rules card.

Engagement definition. Total engagement = likes + comments + shares + link clicks. The main optimisation metric is engagement rate = total engagement / reach. Raw engagement and reach are also reported so that content with a large audience is not automatically treated as more efficient.

Dataset KPI	Value
Posts analysed	250
Total reach	3,708,893
Total engagements	285,095
Overall average engagement rate	7.57%

Main findings

Question	Winner	Result	Sample
Best format for engagement efficiency	Carousel	8.08% average ER	61 posts
Best topic	Founder story	8.11% average ER	34 posts
Best exact posting hour	17:00	9.05% average ER	12 posts
Best broad posting window	12:00-16:59	7.75% average ER	42 posts
Best weekday (bonus)	Wednesday	8.46% average ER	31 posts
Best format for raw scale	Reel	22,449 avg reach / 1,715 avg engagements	78 posts

Recommended test pattern: Founder Story + Carousel + 17:00, with Wednesday as the preferred test day. Because the exact three-factor cell is small, this is a data-led test pattern rather than a permanent causal rule. If an exact time is not practical, the broad 12:00-16:59 window is the best fallback.

2. Method and Verification

2.1 Data preparation

The Posts sheet contains 250 observations with post ID, timestamp, platform, format, topic, caption, reach, likes, comments, shares and link clicks.

A derived Analysis Data sheet was created with total engagement, engagement rate, rounded post hour, daypart and weekday. Rounding the timestamp before extracting the hour avoids floating-point errors such as an intended 17:00 being interpreted as 16:59:59.999.

Pivot-style verification tables were built for format, topic, time window, platform, weekday and exact hour. Both average engagement rate and weighted engagement rate were retained as checks.

Total engagement	Likes + Comments + Shares + Link Clicks
Engagement rate	Total engagement / Reach
Weighted engagement rate	Sum of engagement / Sum of reach

2.2 LLM explanation versus spreadsheet evidence

LLM explanation	Pivot verification	Decision
Carousels may create more dwell/swipe opportunities.	Carousel ER = 8.08% across 61 posts versus 7.57% overall.	Supported
Founder stories may work because they humanise the brand and create a narrative.	Founder story ER = 8.11% across 34 posts, the highest of 8 topics.	Supported
17:00 may align with a strong transition / after-work browsing window.	17:00 ER = 9.05% across 12 posts; strongest exact hour in the data.	Supported as association
Reels should be the best format overall.	Reels win raw volume, but Carousel wins engagement rate.	Partly supported
The best observed three-factor cell should become the permanent rule.	Top cell with at least 5 posts is Carousel + Founder story + Night at 9.47%, but n=5 is too small.	Rejected as overfit

Verification principle: the LLM was used to generate explanations and content ideas, but numerical claims were accepted only after they matched the spreadsheet calculations.

2.3 Improvement rounds documented in the AI-use log

- Round 1 - Raw engagement only: Reels looked like the obvious winner. This was rejected because Reels also had much higher reach.
- Round 2 - Normalised metric: engagement rate showed Carousel as the better efficiency format while Reel remained the scale format.
- Round 3 - Sample-size guard: the strongest three-factor cell used only five posts, so it was not promoted to a permanent rule.
- Round 4 - Timestamp correction: posting hours were rounded before classification, fixing 17:00 records that could otherwise fall into the wrong daypart.

3. Business Recommendation and Content Output

3.1 Posting playbook

Rule	Default	Reason
Format	Carousel	Use when engagement efficiency is the primary KPI.
Topic	Founder story	Lead with a human problem,

Rule	Default	Reason
		decision, trade-off or lesson.
Exact time	17:00	Use as the first timing test; compare against 14:00.
Broad fallback	12:00-16:59	Use when the team needs scheduling flexibility.
Weekday	Wednesday	Prioritise when practical; treat as a bonus finding.
Scale alternative	Reel	Use when reach and total interactions matter more than engagement rate.

Next A/B test. Compare 17:00 (9.05% observed ER, n=12) with 14:00 (8.52%, n=12) using similar format and topic. Collect at least 10 comparable posts per slot before changing the permanent schedule.

3.2 Ten generated posts

The workbook contains 10 AI-assisted Instagram carousel drafts using the same factor pattern: Founder Story + Carousel + 17:00. Each draft contains a product, caption, slide structure, CTA and review status. Products are limited to items already present in the dataset, including Smartwatch, Wireless Earbuds, Air Fryer, Cotton Kurta, Face Serum, Mixer Grinder, Running Shoes and Power Bank.

Product	Opening hook	CTA
Smartwatch	We did not start with a feature list. We started with one question: what should a smartwatch make easier every single day?	Swipe through the journey, then tell us which feature matters most to you.
Wireless Earbuds	Our first earbuds prototype looked finished. It was not. The real work started when we listened to what people disliked.	Which improvement would you notice first?
Power Bank	Power banks look simple from the outside. The decisions behind them are not. Capacity, size, charging behaviour and safety checks all compete for space.	Save this for your next travel-tech checklist.

All 10 captions are marked “AI draft - human review required”. The workbook is the authoritative source for the complete drafts and slide-by-slide structures.

3.3 Working build

Post Optimizer sheet. The workbook includes a simple interactive build with dropdown inputs for format, topic and exact posting hour. It returns the historical engagement rate for each selected factor, lift versus the overall 7.57% baseline, a transparent composite score, and a recommendation. It is intentionally rule-based and explainable rather than presented as a machine-learning forecast.

- Track reach and engagement rate separately.
- Use one clear CTA per post.
- Re-run the analysis after every 20-30 new posts.

- Do not publish AI-generated copy without verifying claims, tone and cultural fit.

4. Responsible AI, Limitations and Reflection

4.1 Limitations

- The dataset is synthetic and contains only 250 posts, so the findings are appropriate for teaching and prototyping but should not be treated as market truth.
- The analysis is observational. Format, topic and time are associated with engagement, but they do not prove causation. Platform algorithm, audience, product, campaign objective, creative quality and season can confound the result.
- The 17:00 recommendation is based on 12 posts. That is enough to justify a test, not enough to guarantee future performance.
- Average engagement rate and weighted engagement rate answer slightly different questions. Both are retained to reduce the risk of drawing conclusions from one statistic.
- AI-generated captions can introduce unsupported product, safety, health or performance claims. Every draft requires human review before publication.

4.2 Responsible-AI controls used

Control	How it was applied
Data privacy	Only the synthetic assignment dataset was analysed; no real personal or confidential data was added.
Verification	Every headline number was checked against spreadsheet calculations.
Disclosure	Generated posts are explicitly labelled as AI-assisted drafts requiring human review.
Transparency	The optimizer is described as a historical heuristic, not as a causal or ML prediction.
Iteration evidence	The AI-use log contains 15 documented entries, including rejected outputs and reasons for changing the approach.

4.3 Reflection

The most important learning from this project was that the first “obvious” answer can be wrong if the metric is poorly chosen. When I looked at raw engagement, Reels appeared to be the clear winner. After normalising by reach, Carousel became the better format for engagement efficiency. That changed the business recommendation and made the difference between scale and efficiency clear.

A second learning was that AI explanations are useful for forming hypotheses but not for deciding whether a pattern is real. The explanations about carousel dwell time, founder-story connection and posting-time behaviour only became acceptable after the spreadsheet pivots supported the numerical pattern. I also had to correct a timestamp precision issue before trusting the posting-time result, which reinforced the need to verify even simple calculations.

If this were deployed with real social data, the next step would be to collect a controlled batch of comparable posts, especially around 17:00 versus 14:00, then rerun the same pivots. I would also add campaign objective, audience segment and creative-quality tags so the analysis can separate the effect of time from the effect of content quality.

5. Conclusion

Final recommendation. Use Founder Story carousels as the primary engagement-efficiency format and test them at 17:00, preferably on Wednesday. Use Reels when the objective is reach or total interaction volume. Treat every result as a hypothesis to be re-measured, not as a permanent social-media law.

Submission package: completed Excel workbook, this report, README for GitHub, and the AI-use log inside the workbook. Before final submission, add the shared ChatGPT conversation link in the evidence column of the AI-use log.

6. Submission Checklist

- Open the completed workbook once in Excel and confirm the Post Optimizer dropdowns recalculate correctly.
- Replace the evidence text in the AI Use Log with the shared ChatGPT conversation URL before submitting.
- Upload the completed workbook, report PDF and README.md to the GitHub repository.
- In GitHub, confirm the repository visibility and copy the repository link for the faculty submission.

Suggested GitHub repository files

File	Purpose
T42_Social_Media_Post_Optimiser_Completed.xlsx	Main working build and all required workbook deliverables
T42_Social_Media_Post_Optimiser_Report.pdf	3-5 page report for submission
README.md	Project summary, findings and instructions for the repository